

DATATALKSCLUB

MLZOOMCAMP 2025

Week 2 notes

Carlos Leal León

October 8, 2025

General Index

1	Car price prediction introduction	4
1.1	Data preparation	4
1.2	Exploratory Data Analysis (EDA)	4
1.3	Log transformation	4
1.4	Missing values	4
1.5	Validation framework	4
1.6	Linear Regression	4
1.7	Base Solution	4
1.8	Root mean squared error (RMSE)	4
1.9	Using the validation split	5
1.10	Simple Feature Engineering	5
1.11	Linear Regression Regularization	5
1.12	Use of the selected model	5
2	Data Preparation	6
3	Exploratory Data Analysis (EDA)	7
4	Validation Framework	8
5	Linear Regression	9
5.1	Baseline	9
6	Root Mean Squared Error (RMSE)	10
7	Feature Engineering	11
8	Regularization	12
9	Model Tuning	13
10	Using the model	13
11	Homework Notebook 02	14
11.1	Exploratory Data Analysis	14
11.2	Partitioning a dataset	14
11.3	Data cleaning	14
11.4	Feature Engineering	14
11.5	Linear Regression and Root Mean Squared Error (RMSE)	15
11.6	Regularized Linear Regression	15
11.7	Using different seeds	15

Figure Index

1	<i>Simple Linear Regression.</i>	9
2	<i>Linear Regression Weights.</i>	9
3	<i>Ridge Regression Regularization.</i>	12

Equation Index

1	Simple Linear Regression	9
2	Simple Linear Regression Weights	9
4	Ridge Regression Regularization	12

1 Car price prediction introduction

The following steps were described briefly as they are covered later in different sections and practiced with examples in my homework notebook[1]. The notebook followed to learn can be found here[2].

1.1 Data preparation

The provided dataset came from a kaggle competition[3] consisting of 11914 observations and 16 features.

Columns were formatted to lowercase and spaces replaced with underscores.

1.2 Exploratory Data Analysis (EDA)

EDA started with the distribution of car prices, during this step it was possible to see how values ranged from 0-200000 in the first visualization. Later a conditional filter was applied and values were concentrated in range 0-100000.

1.3 Log transformation

As car price values are a bit high compared to other values in the dataset the car price was transformed using numpy $\log_{10}$ [4]. This considerably reduced the range of car prices changing the 'shape' of the previously shown distribution.

1.4 Missing values

Now was the turn to look for feature missing values in the dataset. Here there are different options to choose from: Remove missing values, Imputation with different techniques. It must be considered which approach works best for the data you have at hand. In this case imputation of zeros was made.

1.5 Validation framework

In MLZoomcamp this step involves splitting the dataset in train, test, and validation as it is an introduction it is shown how to make it manually to understand the process. Then data is shuffled to select it in a non-sequential way for each split.

1.6 Linear Regression

This step defines a function to train a Linear Regression using a feature matrix and the y values of the train split, the function returns the weights specified as $w[0]$ and $w[1:]$.

1.7 Base Solution

Here numeric features were selected to establish the feature matrix. A dataframe was created using selected features and missing values were filled with zero, after that the feature matrix was initialized with the dataframe values.

Now the Linear Regression was trained and weights were saved in two variables. Predictions were obtained using the weights by adding weight zero to the dot product of the feature matrix and the rest of the weights.

At this point the predictions were plotted to see how close the predicted values are to the training values of the y vector.

1.8 Root mean squared error (RMSE)

A function returns the RMSE subtracting vector of predictions from y vector (for this case the y train vector) then the result is squared and mean is calculated this is the mean squared error, the function returns the square root of the mean squared error.

1.9 Using the validation split

After training the Linear Regression using the training data now the validation data is used to get predictions and calculating the RMSE.

1.10 Simple Feature Engineering

Here the first engineered feature was age obtained by subtracting the year the car was released to the public from the current year. After that a new Linear Regression was trained considering the age feature in the previously selected features.

Later more features were engineered to showcase different examples and different models were trained in the same way.

One important thing to note here is that adding more features does not necessarily increase model performance. I would say that fewer features are better but the key to this statement considers the quality of the selected features i.e. how well do the selected features 'represent' the available data so that the model is able to make an acceptable or optimal prediction.

1.11 Linear Regression Regularization

The trained Linear Regression is now regularized which in simple terms improves the model's generalization performance on unseen data. Remember that the model is always receiving 'unseen' data and if for some reason the model learns the training data so well, that is going to end in poor performance on new data.

To be able to select an appropriate value for regularization different values are tested and compared, to select an optimal regularization value, use the RMSE on each trained model as done previously and compare your results.

1.12 Use of the selected model

After the model comparison and selection it is time to try out the model to make a prediction on car price using an index, you get that index from the test set and train a model by using the set of features that were obtained from the feature engineering. Then you make the prediction and transform the log value back to the original value using `numpy.expml[5]`.

2 Data Preparation

In my eyes preparing data is like getting all your ingredients and make sure they are ready before you go into the kitchen and start cooking. It involves several things, for example: Go an get all the ingredients, clean them, organize them so that they are easy to handle and identify. Once ready you can start manipulating them.

Simple data preparation steps can include:

- Gather data: Data can come from databases, files, or other structured or unstructured sources.
- Read data: Reading csv files is one common way, other options could be xml, json, xlsx or through api calls.
- Familizarize with data: Take a look at a few observations of the dataset.
- Identify features: Check out the feature names and if necessary rename them for smoother use and reference.
- Look at datatypes: Once you familiarize with the dataset observations and features it is easier to infer what type of data you are working with and how it would be more useful, numeric data has certain datatypes and categorical data can be structured in different ways or it may not be necessary at all.

The process highly varies when data is unstructured and features are not well defined, this process requires effort and attention to detail, other cases where data is well defined a structure is easier to identify. You can think of it as a toolbox, but the toolbox may be a good fit for certain features and in some cases you may have to shape or get a different toolbox to structure different kinds of data. In the best scenario data has already been gathered, structured and formatted to an extent, but this is not always the case.

3 Exploratory Data Analysis (EDA)

The purpose of EDA is to not make prior assumptions about the data and understand the underlying structure of the data.

You start investigating the dataset for reasons like:

- **Identify errors:** Observations in a dataset are not error free for whatever reason, so it is important to identify obvious and not so obvious errors (a negative age, wrong dates, misformatted quantities, the list goes on)
- **Detect outliers or anomalies:** There are cases where observations do not fall within what is common or expected such behaviour has a cause and it can be detected using visualizations or descriptive statistics.
- **Uncover patterns:** Data can present to you in the form of patterns, in some cases they are just there but most of the time they have to be discovered as this can help prevent or prepare for certain events ahead of time, for this different types of visualizations can help in the process as it is easier to look at the 'shape' of a feature whenever possible than looking at a table filled with just numbers.
- **Variable Relationship:** A change in certain variables can affect how others behave thus it has an influential effect that can change the model performance. So it is valuable to identify them for later processing.

There are more specific reasons why you conduct EDA, e.g. Class imbalance, insufficient data, poor quality data, outdated data, you get the point now.

EDA consumes most of the time as the effect of understanding and making data ready for use reduces bias, prevents poor model performance, ensures data quality and can help validate a model's assumptions when required.

4 Validation Framework

Once the dataset is ready to be used is time to split it into train, validation and test. For each you need to obtain the feature matrix (X) and vector of target (y).

Steps to split the dataframe are simple:

- **Size:** Define the size of each of the splits, commonly the train set uses most of the data and validation and test split use what is left in half, e.g. (60, 20, 20) or (70, 15, 15) respectively. The more data the better as splits can have more observations to work with.
- **Shuffling:** Observations in the dataset are shuffled to ensure each split contains non-sequential records from the dataset.
- **Create splits:** Each partition is created using the shuffled observations and size of each split.

An example of this can be found in [2] under the Validation framework section.

5 Linear Regression

The objective of a Linear Regression is to adjust a line for the available data and make predictions on unseen data. Recall the input for this model is a Feature matrix (X), the outcome is a vector of predictions ($\hat{y}$) where the values in vector ($\hat{y}$) try to be as close as possible to the real (y) values.

The Linear Regression formula looks like:

$$g(x_i) = w_0 + \sum_{j=1}^n w_j \cdot x_{ij} \quad (1)$$

Figure 1: *Simple Linear Regression.*

Where:

- w_0 : Is the bias term.
- $w_1 \cdot x_{i1} + \dots + w_n \cdot x_{in}$: The sum of each feature value times their corresponding weight.

To obtain predictions as close as possible to the real (y) it is required to calculate the weights from the general Logistic Regression equation. As the feature matrix (X) is not square an **approximate** solution is required. Using the *Gram Matrix* i.e. dot product between feature matrix (X) and its transpose X^T .

The result is the vector of weights also known as coefficients which is the closest possible solution to the equation.

$$w = (X^T X)^{-1} X^T y \quad (2)$$

Figure 2: *Linear Regression Weights.*

Where:

- X : Is the feature matrix.
- X^T : Transpose of feature matrix.
- y : The target values.

Is important that you note the inverse of *Gram Matrix* $(X^T X)^{-1}$ as the goal here is to find the vector that minimizes *the residual sum of squares* in simple terms how far data points are from the adjusted line as what you want is to have them as close as possible to the adjusted line.

5.1 Baseline

Only numerical features are included in the feature matrix. Either from the cleaned or feature engineered dataset (results may vary). Weights are derived from the train dataset in order to obtain the predicted vector w . Once a model is trained both vectors real and predicted (y) can be visualized for comparison. It is called baseline as you are only including the bias term and vector of weights or coefficients.

6 Root Mean Squared Error (RMSE)

The RMSE measures the average difference between a statistical model's predicted values and the actual values. Mathematically it is the standard deviation of the residuals. Residuals represent the distance between the *Regression Line* and the data points. As data points move closer to the regression line the model has less error, thus RMSE is lower. In general a Linear Regression model with lower RMSE produces more precise predictions.

RMSE looks like:

$$RMSE = \sqrt{\frac{1}{m} \sum_{i=1}^m (g(x_i) - y_i)^2} \quad (3)$$

Where:

- $g(x_i)$: Predited value.
- y_i : Real/actual value.
- m : Number of observations in the dataset.

7 Feature Engineering

Raw data is transformed into meaningful features that improve the performance of machine learning models.

You get to select features, create them, modify them (transform), extract other relevant features.

High quality features can enhance model accuracy, speed up training, reduce overfitting.

From my perspective this is an extension of data cleaning as you have to shape your data in a certain way for it to become useful, examples:

- **Extract features:** Often features can be broke down into separate features, e.g. a date made of day, month, year, hour, minute, second.
- **Create a feature:** Involves using one or more available features to make a new one, e.g. getting a value such as age from a date of birth and the current year.
- **Modify/Transform:** Using a feature or group of them you can apply a transformation, e.g. scaling, imputation, replacements, encoding. As presented in the lectures the number of doors a car has appears to be numerical when you can look at it as a categorical feature regarding the number of doors in each car.
- **Selection:** This is most common when there are many features to choose from, when this happens you have to identify features that are significant over others so you can use only the significant features to train the model, e.g. correlation, a model assumption or statistic measure.

This step can become easier as you get experience as not all datasets behave in the same way and some models seem to have certain restrictions to be good candidates while others are more flexible when choosing features.

8 Regularization

The regularization applied in the notebook was *Ridge Regression* also known as *L2 Regularization*. Is particularly used when features are correlated or when the number of predictors is high. Recall that during the feature engineering process several features were created [6]. It consists of adding a factor α to the diagonal of matrix $X^T X$ in order to prevent the values in vector w from growing too large in value either positive or negative.

Ridge Regression looks like:

$$w = (X^T X + \alpha I)^{-1} X^T y \quad (4)$$

Where:

- X : Is the feature matrix.
- X^T : Transpose of feature matrix.
- α : Is the factor.
- I : An identity matrix.
- y : The target values.

Figure 3: *Ridge Regression Regularization*.

Using regularization you can find a solution when there are correlated columns, reduce overfitting and improve model performance. Of course you have to consider the type of regularization as they use different approaches.

9 Model Tuning

In this case the Linear Regularization model tuning was made by regularizing the model trying out different values of the added factor described in *Ridge Regression Regularization* (4) after training different models with different factors and comparing their RMSE of the predicted ($\hat{y}$) and actual (y) values the model with the best RMSE is selected.

10 Using the model

Once the best parameters for the model are found. You have to train a new model using the found parameters. Then use the test split to train the model. Later use the validation split to make predictions and then calculate the RMSE of the predicted ($\hat{y}$) and actual (y) values and finally repeat this last step on the test dataset. At the end you can compare the RMSE obtained to see how well does the model perform.

Now it is time to select an observation from the test split to make a prediction.

The features from the selected observation are saved to a dataframe, then they are prepared (all the cleaning and transformation process) so they are ready to be used by the model for training.

Once ready it is time to use the weights from the trained model and make a prediction using the feature matrix (X) from the selected observation. Notice that if the value of vector (y) has been scaled or transformed e.g. a function like `log1p` was used you have to scale or transform the prediction value back to it's original scale. Now you can review the result from the prediction.

11 Homework Notebook 02

Just mentioning what I read and describe a bit my learning process with this homework.

11.1 Exploratory Data Analysis

In this case you are presented with a question: Does it have a long tail?

First you have to understand what does the long tail mean, so you start searching things like: What does it mean for a dataset feature to have a long tail? or What is a feature with a long tail in machine learning?

And you are introduced to Probability distributions, how they look and some of their characteristics for starters. All of this leads to visualizations in this case histogram plots either from matplotlib or seaborn commonly.

- matplotlib histogram: Find it here [7]
- seaborn histogram: Find it here [8]
- Statistics by Jim: I found this website [9] while searching for plots of probability distributions as I have seen some of them and just looking at the formulas can be intimidating.

I used a combination of Answer with ai (Brave browser) to get a brief introduction, then visit blogs, look for digital books and my own notes from other courses.

11.2 Partitioning a dataset

Commonly this is done using scikit-learn so you skip the process of how it is done. Here I defined a function similar to the one presented in the course materials but I changed how the shuffling was done by using a generator as suggested by the sonar tool. Doing this lets you understand what is done by the library of course in an optimized way.

11.3 Data cleaning

Later you have to deal with missing values filling with zero and with the mean this is straightforward I simply defined a function to do this for each case. I verified there were no more missing values and separate the feature matrix (X) from the vector (y) in each split. Splits consisted of 60%,20%,20% for train, validate, test respectively.

11.4 Feature Engineering

For this part I was not sure exactly how to approach it then I remembered the Does it have a long tail? and I created a few visualizations to see how each feature was distributed, after that in the car price notebook the log1p transformation was applied and I thought of it but instead of applying a log transformation because that was used due to the large values of the price feature. So, I asked What were common feature engineering tasks?

Here I came across techniques like normalization, standardization and log transformation was mentioned again. Knowing I was close I made another question and decided to apply z-score scaling as a way of basic Feature Engineering and test it.

I also created a feature from the year the car was released to the public. It is important to remove irrelevant features as you no longer need them, just do not forget.

Also here I was able to visualize how the imputation technique that you apply affects let's say the 'shape' or distribution of the features that have missing or duplicated values.

11.5 Linear Regression and Root Mean Squared Error (RMSE)

The Linear Regression is straightforward as you have the example from the car price notebook. The only addition here is the use of the RMSE to see how different are the predictions from the real data. Here what is interesting is to see the distribution of the different predictions vs the real values, this reminds you that the solution you get is an approximation.

At first I thought of rounding the RMSE to 3 decimals but doing that does not let you see the changes in it as later you perform Regularization and it seems that there is no change at all.

11.6 Regularized Linear Regression

Here The Linear Regression code is changed to apply Ridge Regression using different factors already provided in the homework. One interesting here to note here is rounding or not your results when you obtain the RMSE as the selected r is different in each case. As the homework asks you to round them to 2 decimals that is what you do but what about other scenario what would be correct?

11.7 Using different seeds

Later you have to train other Linear Regression models using different seed values without applying Regularization and filling missing values with zero. The objective here is to identify how different are the RMSE scores of each seed by calculating their Standard Deviation. If the result is low then one could say that the model is 'stable'.

References

- [1] cleall (github). "hw notebook" [Online], at: hw-two [Created on October 2025].
- [2] alexeygrigorev (github). "02-carprice" [Online], at: Car price notebook [Consulted on October 2025].
- [3] Cooperunion Jeffrey Shih. "Car Features and MSRP" [Online], at: car-csv [Consulted on October 2025].
- [4] Numpy 2.3. "numpy.log1p" [Online], at: log1p [Consulted on October 2025].
- [5] Numpy 2.3. "numpy.expm1" [Online], at: expm1 [Consulted on October 2025].
- [6] Ksenia Legostay. "Regularization in Regression" [Online], at: regularization [Created on October 2025].
- [7] Matplotlib Org. "hist" [Online], at: hist [Created on October 2025].
- [8] Seaborn Pydata Org. "histplot" [Online], at: histplot [Created on October 2025].
- [9] StatisticsByJim. "Probability Distributions" [Online], at: distributions [Created on October 2025].